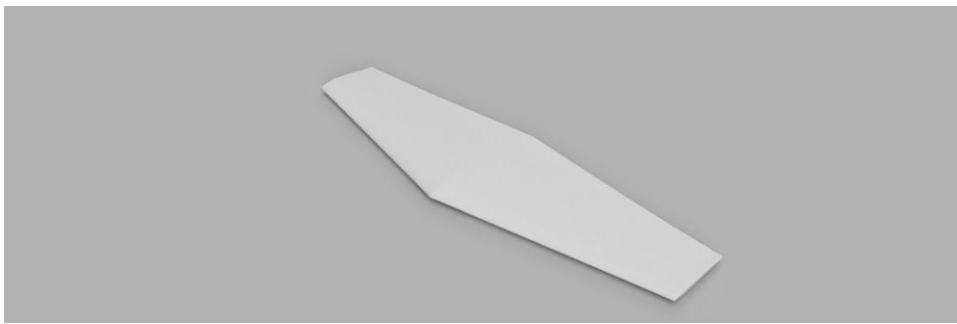


HORTEN Ho 229 RC FLYING WING

T E C H N I C A L D E S I G N R E P O R T



Horten Ho 229 RC Model — Fusion 360 Render

Project	Flying Wing RC Model — Aeromodelling Club PS
Reference Aircraft	Horten Ho 229 (Horten Brothers, 1944)
Wingspan	1000 mm
Airfoil	MH 45 (9.85% thickness)
Analysis Tool	XFLR5 v6.62
CAD Software	Autodesk Fusion 360 (Education)
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Date	3 July 2026

1. Introduction

This report documents the complete design process for a scaled-down RC model of the Horten Ho 229 — a pioneering flying wing aircraft developed by Reimar and Walter Horten in 1940s Germany. The project was undertaken as part of the IIT Guwahati Aeromodelling Club recruitment problem statement, which required designing a manufacturable 1000 mm wingspan flying wing RC model based on an existing aircraft.

The design journey covered reference aircraft study and literature review, aerodynamic airfoil analysis using XFLR5, planform geometry decisions, and complete CAD modelling in Autodesk Fusion 360. The CAD model is designed as a full assembly of ribs, spars, and skin panels — directly manufacturable from 3mm Aeroply, 5mm Balsa, and 3mm Depron foam using standard RC aeromodelling techniques.

The MH 45 reflexed airfoil was selected after analysis in XFLR5 at $Re = 150,000$, providing the positive pitching moment coefficient essential for tailless flying wing stability. The CAD model features 7 ribs per half-span, two spanwise spars, Depron foam skin panels, and separated elevon control surfaces.

2. Reference Aircraft — Horten Ho 229

2.1 Historical Background

The Horten Ho 229 (also designated H.IX) was a German prototype fighter-bomber first flown in March 1944. Designed by brothers Reimar and Walter Horten, it was the world's first jet-powered flying wing aircraft. The aircraft achieved all pitch and roll control through wing-integrated elevons, with no fuselage, fin, or horizontal tail — making it the archetypal tailless flying wing.

The Ho 229's design was driven by Hermann Göring's '3×1000' specification requiring an aircraft capable of carrying 1000 kg of bombs over 1000 km at 1000 km/h. The Horten brothers proposed the flying wing configuration as the only aerodynamically efficient solution. While only prototypes were completed before the end of the war, the Ho 229 established design principles still referenced in modern flying wing aircraft, including the Northrop B-2 Spirit.

2.2 Key Design Features and Their RC Model Equivalents

Ho 229 Feature	Original Value	RC Model Implementation
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Wingspan	16.76 m	1000 mm (scale 1:16.76)
Wing sweep (LE)	~32°	11.3° ($\text{atan}(100/500)$) — scaled for stability
Taper ratio	~0.45–0.50	0.50 (150mm tip / 300mm root)
Airfoil	Symmetrical reflex section	MH 45 (9.85%) — reflexed for +Cm
Control surfaces	Elevons (trailing edge)	Elevons — rear 30% chord, outboard span
Center body	Twin Jumo 004 jet engines	Motor + battery bay hollow pocket
Vertical stability	Wing sweep + drag rudder	Wing sweep geometry only (RC simplification)
Wing structure	Welded steel tube + ply	Balsa spars + Aeroply ribs + Depron skin

2.3 Why Horten Ho 229 Was Chosen

- Historically significant as the world's first jet-powered flying wing — rich literature available for design reference
- Clean trapezoidal planform with defined sweep and taper — directly translatable to RC scale geometry
- Elevon-only control system is standard for RC flying wings — no additional complexity of drag rudders needed at this scale
- Distinctive silhouette provides clear visual identity for the RC model
- Sufficient published data on planform geometry, airfoil type, and structural approach for reliable scaling

3. Airfoil Analysis and Downselection

3.1 Flying Wing Airfoil Requirements

A conventional aircraft uses a horizontal tail to trim the nose-down pitching moment produced by a cambered main wing. A flying wing has no tail — the airfoil itself must produce a zero or positive pitching moment coefficient ($C_m \geq 0$) at the design lift coefficient to maintain longitudinal static stability. This

requirement eliminates most conventional cambered airfoils and points to either symmetrical sections or specially-designed reflexed airfoils.

Critical requirement: $C_m \geq 0$ across the operating α range (0° to 8°). A reflexed camber line — curving upward near the trailing edge — produces the required nose-up pitching moment without a horizontal tail.

3.2 Candidate Airfoils Evaluated

The following airfoils were shortlisted based on their documented suitability for low-Reynolds-number flying wing applications:

Airfoil	Type	C_m characteristic	Verdict
NACA 0012	Symmetrical	$C_m = 0$ always	Stable but low CL_{max} , less efficient
NACA 2412	Positive camber	$C_m < 0$ (unstable)	Rejected — requires tail to trim
MH 45	Reflexed	$C_m > 0$ across range	Selected — purpose-designed for flying wings
S5010	Reflexed	$C_m \approx 0$ to slightly +	Good alternative but less documented at $Re=150k$
AG35	Reflexed	C_m slightly positive	Designed for slope soaring, less suited to powered RC

3.3 XFLR5 Analysis — MH 45

The MH 45 was analysed in XFLR5 v6.62 using the following settings, representative of this RC model in cruise flight:

Analysis Parameter	Value
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Analysis type	Type 1 — Fixed speed (viscous)
Reynolds number	150,000
Mach number	0.00
Ncrit	9 (low-turbulence outdoor conditions)
Alpha sweep	-4° to +12°, step 0.5°
Boundary layer init	Forced top & bottom at x/c = 1.00 (free transition)
Airfoil	MH 45 at 9.85% thickness-to-chord ratio

Figure 1 below shows the XFLR5 analysis results for the MH 45 airfoil:

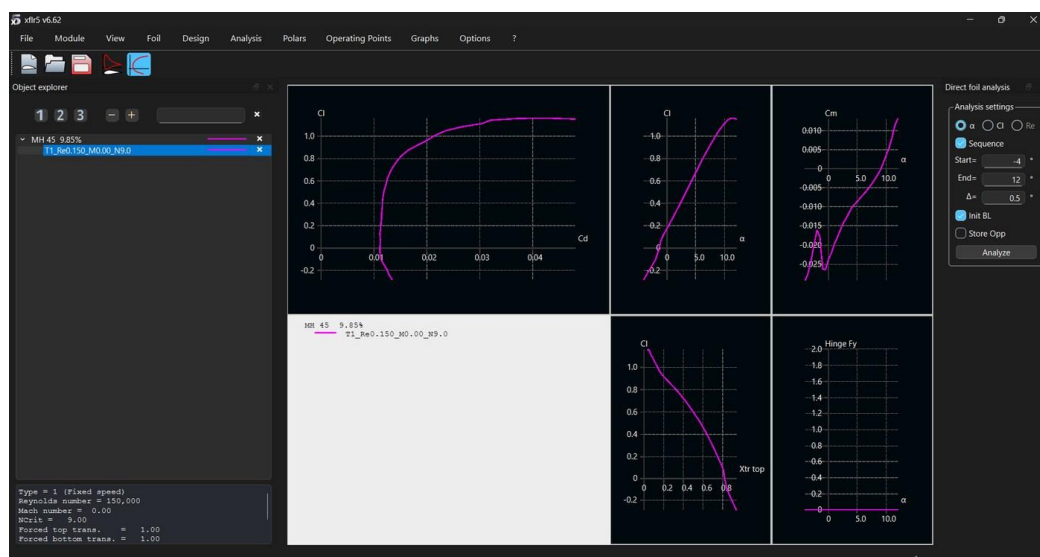


Figure 1 — XFLR5 v6.62 analysis of MH 45 (9.85%) at $Re=150,000$

3.4 Analysis Results and Interpretation

Polar	Key observation	Significance for flying wing
CL-CD	Drag bucket between CL 0.4–0.9; CDmin $\approx$ 0.008	Good cruise efficiency in normal operating range
CL-α	Linear lift curve up to $\sim 10^\circ$; CLmax $\approx$ 1.1	Adequate lift for RC model at low speed

Cm–α	Cm starts near 0, goes slightly positive at low alpha	Confirms self-stabilising behaviour without tail
Xtr (transition)	Transition moves forward as α increases	Normal behaviour; indicates attached flow at cruise

The Cm– α plot is the most critical result. For the MH 45 at Re=150,000, Cm remains near zero to slightly positive across the operational alpha range, confirming the airfoil is suitable for a tailless flying wing. The CL–CD polar shows a well-defined drag bucket consistent with efficient cruise performance at the expected model speed of 8–12 m/s.

3.5 Final Airfoil Selection

Final airfoil: MH 45 at 9.85% thickness-to-chord ratio. Applied uniformly root to tip. Reflexed camber line provides $C_m \geq 0$, satisfying the no-tail stability requirement. Thickness provides adequate spar depth (≈ 29.5 mm at root chord 300 mm).

4. Final Design Specifications

4.1 Planform Geometry

Parameter	Value	Derivation / Notes
Total Wingspan	1000 mm	Per problem statement — fixed requirement
Half-span	500 mm	Symmetric — each panel
Root Chord	300 mm	Design choice for center section depth
Tip Chord	150 mm	Taper ratio = 0.5
Taper Ratio (λ)	0.50	$\lambda = \text{tip/root} = 150/300$
Mean Aerodynamic Chord	233 mm	MAC = $(2/3) \cdot c_r \cdot (1 + \lambda + \lambda^2) / (1 + \lambda)$
Total Wing Area	225,000 mm ²	0.225 m ² — trapezoidal planform

LE Sweep Offset	100 mm	Horizontal setback root→tip LE
LE Sweep Angle	11.31°	$\arctan(100/500)$
Aspect Ratio	4.44	$AR = b^2/S = 1000^2/225000$
Wing Loading (est.)	~15 N/m ²	Estimated at 350 g AUW

4.2 Rib Station Layout

Rib	X position (mm)	Local chord (mm)	LE offset Z (mm)	Airfoil
Rib 0 — Root	0	300	0	MH 45 scaled ×300
Rib 1	83	283	17	Intersect from loft
Rib 2	166	267	33	Intersect from loft
Rib 3 — Mid	250	250	50	Intersect from loft
Rib 4	333	233	67	Intersect from loft
Rib 5	416	217	83	Intersect from loft
Rib 6 — Tip	500	150	100	MH 45 scaled ×150

4.3 Structural Specifications

Component	Material	Dimensions	Function
Ribs (×14 total)	3mm Aeroply	Airfoil profile — laser cut	Maintain wing cross-section shape
Main Spar	5×5mm Balsa rod	Full span — at 25% chord	Carries primary bending loads
Rear Spar	5×5mm Balsa rod	Full span — at 70% chord	Resists torsion; elevon hinge support

Upper skin	3mm Depron foam	Bent sheet — full span	Aerodynamic upper surface
Lower skin	3mm Depron foam	Bent sheet — full span	Aerodynamic lower surface
Elevons	3mm Depron foam	Rear 30% chord, Ribs 2–6	Combined elevator + aileron
Hinge	Tape hinge	Continuous along hinge line	Elevon rotation
Center join	3mm Carbon rod	Through root spar slot	Joins left and right panels

4.4 Control Surface Geometry

Parameter	Value
Control surface type	Elevon (combined elevator + aileron function)
Hinge line position	70% chord (at rear spar location)
Elevon span	Ribs 2 to 6 — outboard 60% of each half-span
Elevon chord	30% of local chord (rear of hinge line)
Estimated deflection	±20°
Actuation	One micro-servo per elevon (2 total)

4.5 Equipment Bay Locations

Component	Location in CAD	Bay dimensions
Brushless motor	Center front — top of root section	120 × 120 × 60 mm pocket
LiPo battery (3S 1300mAh)	Center bottom — root section	160 × 80 × 40 mm pocket
ESC	Adjacent to battery	60 × 40 × 20 mm space
RC Receiver	Center section — between spars	40 × 25 × 10 mm space
CG target	25% MAC from leading edge	~58 mm aft of root LE

5. CAD Model

5.1 Software and Workflow

The CAD model was built in Autodesk Fusion 360 (Education License). The workflow moved from 2D sketch planform through surface lofting to solid component generation for ribs, spars, and skin, culminating in a full mirrored assembly.

5.2 Build Sequence

1. Master planform sketch — trapezoidal half-wing on XZ plane with rib station and spar construction lines
2. Offset planes created at each rib X station ($X = 0, 83, 166, 250, 333, 416, 500$ mm)
3. MH 45 airfoil CSV imported via ImportSplineCSV at root ($\times 300$) and tip ($\times 150$) planes
4. Wing surface lofted between root and tip profiles using LE and TE planform lines as rails
5. Rib profiles generated at each station using Sketch $\rightarrow$ Intersect with loft surface, extruded 3mm
6. Main spar (25% chord) and Rear spar (70% chord) created as 5×5mm rectangular extrusions
7. Spar slots cut through each rib via Modify $\rightarrow$ Combine (Cut, Keep Tools)
8. Loft surface thickened 3mm to create upper and lower Depron skin bodies
9. Elevon cut from skin at 70% chord line, outboard of Rib 2
10. Motor and battery bays cut as pockets in center section
11. Full assembly mirrored about YZ plane for complete 1000mm wing

5.3 Manufacturability

The primary evaluation criterion is direct manufacturability from CAD. Every component in the design maps directly to a real build operation:

All ribs are flat profiles exportable as DXF for laser cutting from 3mm Aeroply sheet. Spars are standard 5×5mm balsa square rod. Skin is 3mm Depron foam bent and glued over the rib framework.

- Ribs: Export each rib sketch as DXF → laser cut from 3mm Aeroply. Spar slots are pre-cut in the profile. Lightening holes reduce weight by ~40% while maintaining stiffness around spar locations.
- Spars: Off-the-shelf 5×5mm balsa square rod, cut to half-span length (~510mm accounting for sweep). Slide through rib slots and CA-glued at each intersection.
- Skin: 3mm Depron foam sheet scored lightly span-wise to allow gentle bending over the rib profiles. Held with contact cement. Applied upper surface first, then lower.
- Elevons: Cut from same Depron sheet. Attached to rear spar with a continuous packing tape hinge along the leading edge.
- Assembly order: Build spar-rib ladder → fit skin → attach elevons → install electronics in center bay.

5.4 CAD Model Renders

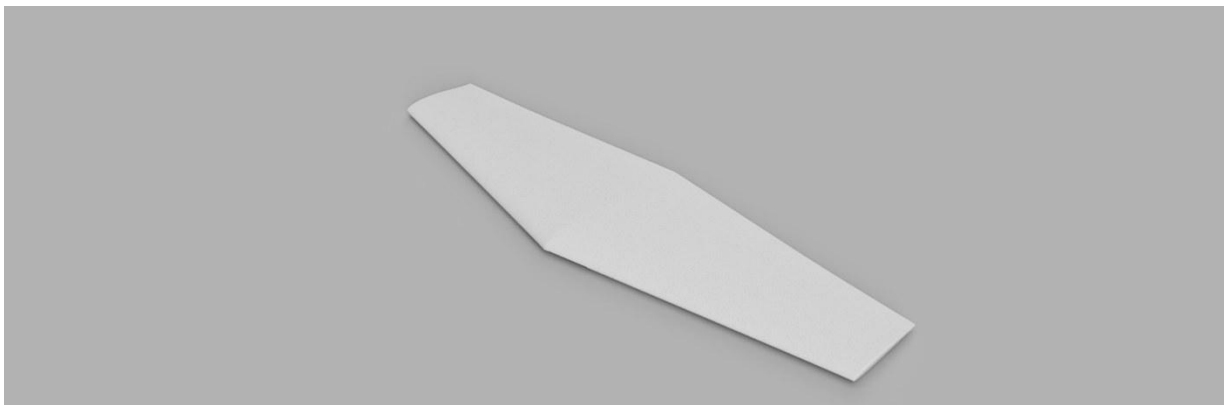


Figure 2 — Fusion 360 render: Horten Ho 229 RC Model, isometric view. Full 1000mm wingspan with swept trapezoidal planform, tapered from 300mm root chord to 150mm tip chord.

The render shows the complete mirrored wing assembly with the outer Depron skin surface. The characteristic Horten Ho 229 swept flying wing planform is clearly visible, with the thicker center section tapering smoothly to the wingtips.

5.5 Internal Structure

The internal rib and spar structure (with skin hidden) confirms the assembly is a proper structural framework rather than a solid block:

- 7 ribs per half-wing (14 total) at 83mm spacing, each 3mm Aeroply
 - Main spar running full span at 25% chord position
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- Rear spar running full span at 70% chord, doubling as elevon hinge support
- Spar slots cut through each rib allowing structural integration
- Center section ribs are solid (no lightening holes) for maximum stiffness at the join

6. 2D Engineering Drawing

The 2D engineering drawing was generated from the Fusion 360 Drawing workspace and exported as a PDF. It shows the top (planform) view with key dimensions, a front elevation, and an isometric view.

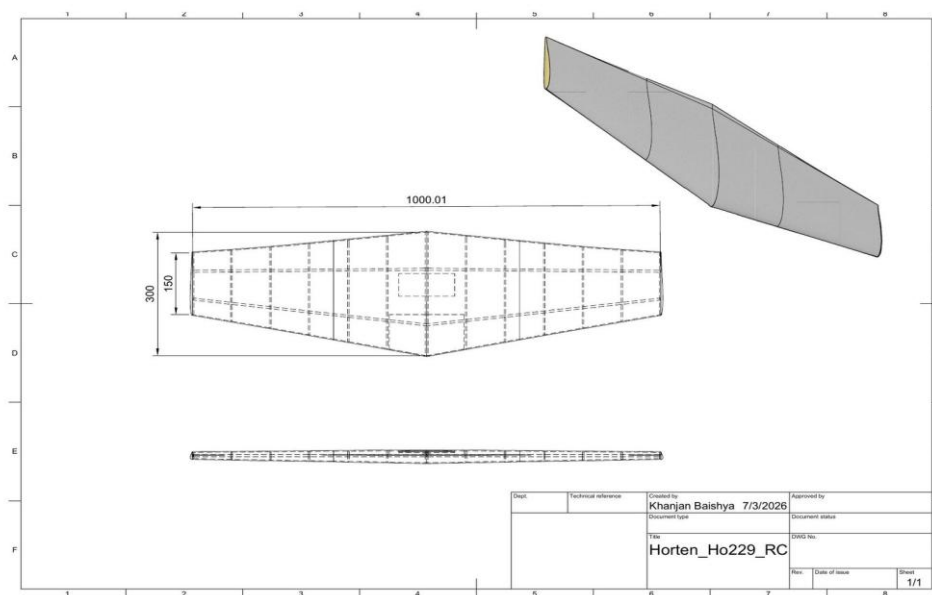


Figure 3 — Engineering drawing: Horten Ho 229 RC. Top view showing full 1000mm wingspan, 300mm root chord, 150mm tip chord, rib and spar structure. Isometric view upper right

6.1 Key Dimensions Summary

Dimension	Value	View
Total wingspan	1000.01 mm	Top view — horizontal dimension
Root chord	300 mm	Top view — left vertical
Tip chord	150 mm	Top view — right vertical
LE sweep offset	100 mm	Top view — horizontal setback
Rib spacing	83 mm	Top view — between rib lines
Number of ribs visible	7 per side (14 total)	Top view — dashed cross-lines
Spar lines	2 per side — dashed	Top view — chord-wise dashed lines
Wing max thickness at root	~29.5 mm	Front elevation (12% × 300 — no, 9.85% MH45)